

Instance Segmentation of Densely Packed Plant Cells under Sparse Annotation in X-ray μ CT

Naoki Yagihara¹, Yuzuko Utsumi¹, Daisuke Yamauchi², Yoshinobu Mineyuki²,
Ichirou Karahara³, Masato Hoshino⁴, Kentaro Uesugi⁴, Yoshio Suzuki⁵,
Akihisa Takeuchi⁴, and Masakazu Iwamura¹

¹Osaka Metropolitan University ²University of Hyogo ³University of Toyama
⁴JASRI, SPring-8 ⁵The University of Tokyo
emailsb26519e@st.omu.ac.jp

Abstract. Unstained X-ray μ CT can capture plant tissue non-destructively at the single-cell level. However, its cell-wall contrast is low, and the cells are densely packed. These conditions make cell instance segmentation difficult. We build on the flow-based method cpsam and introduce two components: z-consensus boundary recovery (ZBR), which strengthens weak boundaries by combining information across z -slices and synthetic training data, which improves label efficiency. Together, these components improve segmentation accuracy under sparse annotation.

Keywords: Cell instance segmentation · Sparse annotation · X-ray μ CT

1 Introduction

The morphology and function of plants are defined by the shape and spatial arrangement of their cells. Quantifying their three-dimensional (3D) morphology is therefore important. Manually segmenting countless cells is impractical, which makes automated instance segmentation necessary. X-ray micro-CT (μ CT) captures an entire tissue without staining and non-destructively. However, its cell-wall contrast is extremely low, and dense packing makes adjacent cells prone to merging. Fig. 1 shows a cross-section of cells in unstained μ CT (A) and the manually segmented ground-truth labels (B). The difficulties are twofold. First, the

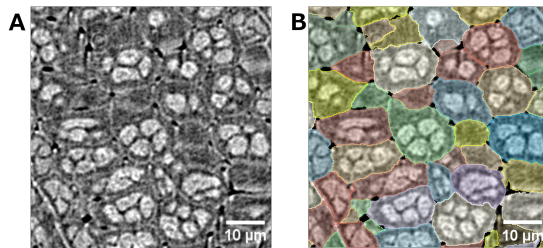


Fig. 1: Plant cells in unstained μ CT (A), and their cell-instance ground-truth (B).

unclear boundaries cause adjacent cells to merge. Second, massive annotation is impractical. The model must therefore generalize from a few annotated cells to the many unannotated cells in the volume.

Cell instance segmentation has three approaches: boundary-based (U-Net [3]), shape-based (StarDist [4,5]), and flow-field-based (Cellpose [6]). Their effectiveness on unstained μ CT, however, remains unclear. We first compare these approaches and show that the flow approach, which does not directly detect boundaries, is the most effective. We therefore build on the flow-based cpsam and add two components. z-consensus boundary recovery (ZBR) aggregates boundary information across z-slices to redraw the boundaries. Synthetic training data mixes synthetic densely packed slices into the flow supervision to improve label efficiency. Our contributions are twofold. First, we show that the flow approach is the most effective on unstained μ CT. Second, we improve segmentation accuracy from sparse annotation with ZBR and synthetic training data.

2 Proposed Method

Figure. 2 gives an overview of the proposed method. The proposed method builds on the flow-field-based Cellpose-SAM (cpsam) [11] and adds two components: z-consensus boundary recovery (ZBR) and synthetic training data. For each pixel, cpsam predicts a gradient toward its cell center. Pixels that converge to the same center form one cell. Because cpsam does not detect boundaries directly, it separates cells better than boundary-based methods, even in unstained μ CT. To handle sparse annotation, we input the whole cross-section, but use only the annotated pixels for training. This setting is shown in Fig. 2(A). The remaining pixels are excluded from the loss. This prevents the model from treating unannotated cells as background and preserves the surrounding context.

At low-contrast boundaries, the flow of cpsam still tends to merge adjacent cells. ZBR corrects these merging errors at the output stage, using the predicted boundaries rather than the input, as illustrated in Fig. 2(B). First, a 5-layer CNN produces a boundary probability map for each slice. The slices around the central slice are then aligned by optical flow and averaged. This reinforces boundaries that are faint in a single slice and suppresses spurious ones. Because boundaries are hard to draw at the pixel level under low contrast, we recast boundary drawing as a fragment-merging problem. The image is first partitioned into many small fragments. A learned classifier then decides, for each pair of neighboring fragments, whether they belong to the same cell, and merges matching ones.

Synthetic training data supplements the annotations to improve label efficiency. We generate a dense structure with no intercellular spaces using a 3D weighted Voronoi partition. Boundary and interior intensities are taken from real data, and texture and noise are added. A key step is matching: we match the visibility of the synthetic boundaries to that of the real data. Boundaries that are too sharp do not transfer to real images, so this matching is essential. The matched cross-sections are not used for pre-training. Instead, they are fed alternately with the annotated cells during training.

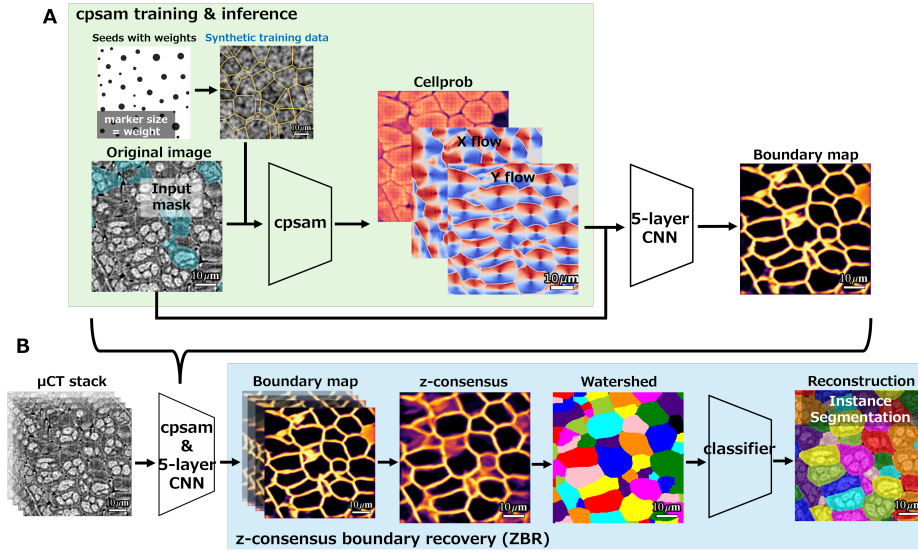


Fig. 2: Overview of the proposed method. (A) Per-slice boundary prediction. First, cpsam predicts a cell probability map and X/Y flow. A 5-layer CNN then regresses a boundary map from them. We train cpsam from a few annotated cells, with synthetic data mixed in. (B) z-consensus boundary recovery (ZBR). Boundary maps across a μ CT stack are first aligned and averaged (z-consensus). The averaged map is then over-segmented by watershed into fragments. Finally, a learned classifier merges same-cell fragments to produce the instance segmentation.

3 Experiments

We captured *Arabidopsis thaliana* seeds at beamline BL20XU of SPring-8. This produced an unstained μ CT volume with an effective pixel size of $0.2428 \mu\text{m}$. From this volume, we selected a region of $256 \times 256 \times 64$ voxels. We then labeled every cell manually, using the unstained image alone, to obtain the ground truth. For each cross-section, only the masks of k cells are used for training, and the remaining cells are used for evaluation. We vary k over $\{1, 5, 10, 20\}$. For evaluation, we use two instance-level metrics: the Aggregated Jaccard Index (AJI) [13] and Panoptic Quality (PQ) [14]. AJI penalizes merging, splitting, boundary displacement, and over-segmentation together. PQ measures both the overlap quality of matched pairs and the number of false positives and misses.

4 Results

Figure 3 (A) compares the three approaches under the same sparse-annotation condition. This comparison shows how the cell representation affects segmentation accuracy on unstained μ CT. We compare the boundary approach (U-Net [3], PlantSeg [8]), the shape approach (StarDist [4]), and the flow approach (cpsam

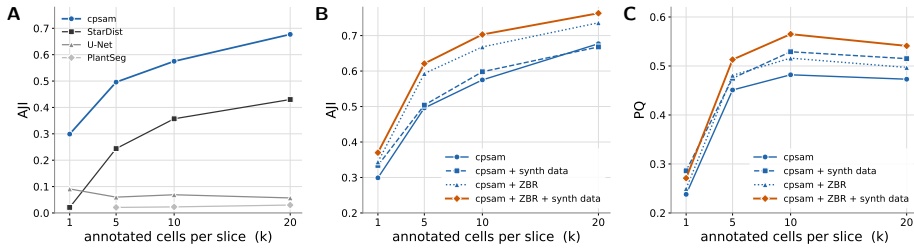


Fig. 3: Segmentation accuracy versus the number of annotated cells per slice (k). (A) Comparison of segmentation approaches in AJI. The flow-based cpsam outperforms the shape-based StarDist, as well as the boundary-based U-Net and PlantSeg. (B) AJI and (C) PQ of cpsam with z-consensus boundary recovery (ZBR) and synthetic data. The two components are added individually and in combination. The combination performs best at every k .

[11]). The flow approach performed best, the shape approach was in the middle, and the boundary approach performed worst. In the boundary approach, the low contrast blurred the predicted boundaries, so segmentation barely held. The shape approach assumes a simple star-convex shape. This assumption does not fit plant cells, which deform in diverse ways under dense packing, so it stayed in the middle. The flow approach depends less on boundaries, which makes it the most effective on this data.

Figure 3 (B) and (C) show the contributions of z-consensus boundary recovery (ZBR) and synthetic training data. We added them to the flow foundation cpsam, both individually and in combination. For AJI, the combination performed best at every k . For PQ, it performed best at all k except $k = 1$. The combination improved AJI by up to about +0.12 and PQ by up to about +0.08. In most cases, each component alone also outperformed the foundation, and the two act complementarily. Boundary recovery mainly raises boundary quality, which improves AJI and PQ. Synthetic training data improves label efficiency when few annotations are available. PQ drops slightly at $k = 20$. As fewer cells remain for evaluation, the predicted count does not decrease accordingly, leaving over-segmentation that raises false positives and lowers PQ. AJI is less affected, and this drop might be reduced by tuning the agglomeration threshold per k .

5 Conclusion

We addressed cell instance segmentation from sparse annotation. Our target was *Arabidopsis thaliana* seeds in unstained X-ray μ CT. We first compared three approaches and found that the flow approach is the most effective. Building on this approach, we introduced two components. The first is z-consensus boundary recovery (ZBR), which raises boundary quality. The second is synthetic training data, which improves label efficiency. Together, these components consistently improve segmentation accuracy.

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